Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. Calculate the mean of 2, 4, 6.
2. Check whether $(\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$ can be a distribution.
3. Calculate $(2-5)^2$.
4. If $P(A) = 0.6$, calculate $P(\overline{A})$.
5. A gross gain of 8 units and a stake of 3 give what net gain?
6. Calculate $\sqrt{0.25}$.

After the diagnostic. Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

Speaking and logic. Explain why a theoretical mean can be an impossible gain. Present the difference between a distribution, a simulation and a data set.

Investigate before formalising

Create two games with three equally likely outcomes, both with expectation 0 but different standard deviations. Check their distributions and explain the difference.

Organisation. Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.